PrintablePrograms.png: Byte-Exact Recovery of Programs from Physically Printed Pages

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July 27, 2026

ABSTRACT. The dominant carriers of software are invisible: magnetic domains, charge wells, and radio packets. We present PrintablePrograms.png, a toolkit that makes the printed page a first-class program carrier. A program is typeset legibly in the middle of a page and simultaneously encoded, byte-exactly, into a thin frame of saturated color cells around the page edge. The frame self-describes its geometry, physical scale, color palette, and network origin, so a screenshot, scan, or photograph of the page suffices to reconstruct the program with no out-of-band knowledge. We describe the format, an eight-color modulation scheme with per-print calibration, a CRC-guarded repetition-coded packet layout, and a family of format converters that route every representation through a canonical XML document carrying both byte-exact source and a structural AST. This paper is itself a demonstration: its canonical XML source builds the PDF, HTML, and Markdown editions you may be reading, and each printed page of the PDF carries this document in its own data frame.

Keywords: printable programs, visual codes, fiducial markers, paper computing, Tcl, program archival

1 INTRODUCTION

Programs on paper have a long history: type-in listings in hobbyist magazines, punched cards, and archival microfilm all treated paper as a distribution medium. Paper is durable, human-inspectable, self-powered, and survives institutional collapse better than any spinning or charged medium. What paper lost, when software grew, was byte-exactness: retyping a listing introduces errors, and OCR of source code remains unreliable precisely where it matters (identifiers, punctuation, whitespace).

PrintablePrograms.png restores byte-exactness while keeping legibility. Every printed page carries two synchronized channels: a human channel (the program text, typeset at a minimum physical size of 12pt) and a machine channel (a border band of large color cells that encodes the program plus enough metadata to decode itself). The two channels cover for each other: the machine channel makes recovery exact; the human channel makes the artifact auditable, quotable, and partially recoverable even when the machine channel is damaged.

The design descends from a lineage of physical-computing systems: the DigitalDesk's insistence that paper and computation share a surface

[Wellner 1993], tangible interfaces [Ishii and Ullmer 1997], Weiser's vision of computation receding into the environment [Weiser 1991], and the communal computing rooms of Dynamicland [Victor et al. 2017] and Folk Computer [Rizwan and Michaud-Agrawal 2023], where programs are literally pieces of paper on tables. Those systems locate programs by fiducial tags and fetch the source over a network; a PrintablePrograms page needs no network, though it names one as a fallback.

2 RELATED WORK

Two-dimensional visual codes are mature technology: QR codes [ISO/IEC 18004 2015] and Data Matrix reach kilobyte capacities with strong Reed-Solomon error correction [Reed and Solomon 1960], and robust fiducial systems such as ARToolKit [Kato and Billinghurst 1999] and AprilTag [Olson 2011] solve detection and pose. PaperBack [Yuschuk 2007] demonstrated dense full-page archival backup (hundreds of kilobytes per page) at the cost of legibility: the page becomes a uniform gray texture. Literate programming [Knuth 1984] made the printed program a publication artifact but kept recovery manual.

PrintablePrograms.png occupies a deliberate middle point: the page remains, at a glance, a document -- title, code, margins -- and the machine channel is confined to a decorative-looking border. Capacity is sacrificed (kilobytes, not megabytes) for a page that a human would pin to a wall. The frame is self-describing in a way that dense archival formats are not: physical dimensions, cell pitch, palette, and origin URL are all recovered from the pixels themselves.

3 FORMAT DESIGN

A page is a cell grid. A quiet white margin (for printer comfort; detection does not require it) surrounds a band of color cells; the interior holds the typeset program. Cells modulate three bits each through an eight-color palette at the corners of the RGB cube -- black, red, green, blue, cyan, magenta, yellow, white -- the colors maximally separated under the subtractive gamut of consumer inkjet inks. QR-style finder fiducials occupy all four corners. Beside the top-left fiducial sit a calibration strip (the eight palette colors in fixed order, letting the decoder re-learn the palette per print) and three self-description words: band thickness, grid width, and grid height, each 24 bits. Self-encoding the grid dimensions makes cell recovery immune to accumulated pitch error, which defeats run-length-only estimation on wide grids.

The band payload is a single packet: magic, version, a header of key=value lines (hostname, IP, program identifier, a UTC-millisecond filename, page and cell dimensions in millimeters, creation time, origin URL), the leading bytes of the program, and a CRC32. The packet repeats to fill the band; a decoder accepts the first copy whose CRC verifies. Because physical dimensions ride in the header, every artifact encodes its own absolute scale: a decoder reports pixels per millimeter for any raster it is handed, and US Letter editions of this paper are emitted with fixed 8.5 by 11 inch geometry declared in the frame itself.

When a program exceeds band capacity the frame carries a prefix, a complete=0 flag, and the origin URL from which the full program can be fetched (http://ip/folk-data/program/filename); the typeset human channel carries the rest of the meaning.

4 TOOLCHAIN

The toolkit is implemented in Tcl [Ousterhout 1990], the language of the Folk Computer ecosystem it serves. A canonical XML document is the hub of all conversions: it carries the byte-exact source (base64), a per-command breakdown, and a full abstract syntax tree with typed words and nested scripts, from which a semantically equivalent program can be regenerated. The press (bin/press.tcl) converts any pair among .tcl, .xml, .png, .jpg; raster carriers hold the XML in PNG tEXt chunks and JPEG COM segments, so a lossy JPEG still round-trips the source byte-exactly. The printout tool (bin/printout.tcl) emits and decodes page frames; the decoder assumes only an axis-aligned raster and tolerates anisotropic rescaling, blur, and recompression. Perspective and rotation rectification are delegated to the surrounding computer-vision layer, as in AprilTag-based rooms.

The white paper tool (bin/whitepaper.tcl) is the toolkit eating its own cooking: it typesets this document from WhitePaper.xml into US Letter page frames set in the Libertinus revival of Linux Libertine -- the ACM's standard typeface family [Boldt et al. 2012; ACM 2017] -- and emits every published edition, including carrier programs in folk, Rust, Go, and C++ that reproduce the PDF, images, diagrams, and formatting from constants embedded in their own source.

5 EVALUATION

The test suites hold 34 checks. Digital round trips are byte-exact across every format pair, including chains through lossy JPEG. Physical-channel simulations decode byte-exactly after 137 percent upscale, 63 percent downscale, and a rescan simulation of 80 percent rescale with Gaussian blur and JPEG quality 80. A committed sample (storyboard.printout.rescan.jpg) demonstrates recovery from degraded pixels using no metadata. Capacity at default geometry (2mm cells, 12-cell band) is roughly 1.7 kilobytes per US Letter page, sufficient for the programs the Folk community actually prints; larger programs degrade gracefully to prefix plus origin.

6 LIMITATIONS AND FUTURE WORK

The decoder requires an axis-aligned image; camera-in-the-wild decoding needs the perspective rectification that fiducial systems already provide. Error correction is repetition plus CRC; Reed-Solomon coding would multiply usable capacity under damage. The sketch that motivated this work calls for burn-blend barcode layers encoding six degrees of freedom of pose and scale per storyboard frame, paper-size preset families, and transpiled carriers (.tk.folk, .c.folk, .folk.rs, .folk.go); the mm-parameterized geometry and the carrier generators in this paper are the first steps of that roadmap.

7 ACKNOWLEDGMENTS

To the Folk Computer community, for insisting that programs belong on tables, and to a hand-drawn orange-marker system diagram, for being a better specification than most specifications.

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# storyboard.tcl -- a folk-flavored sample: fake 6DOF frame descriptions
# (echoes the sketch: barcodes encoding 6DOF position + scale per frame)
proc frame {id pos rot scale} {
    puts "frame $id @ pos=$pos rot=$rot scale=$scale"
}

frame 1 {0.0 0.0 0.0}   {0 0 0}    1.0
frame 2 {0.1 0.0 0.25}  {0 15 0}   1.0
frame 3 {0.2 0.05 0.5}  {0 30 5}   0.8

puts "storyboard: 3 frames, src (PNG)"
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Figure 1. A printable program page: minimal color-cell frame, code at 12pt, byte-exact recovery from pixe